SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards ***(BL2, BL6)***

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**

Annexure A

Coding challenge

Code of Conduct:

I **M Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

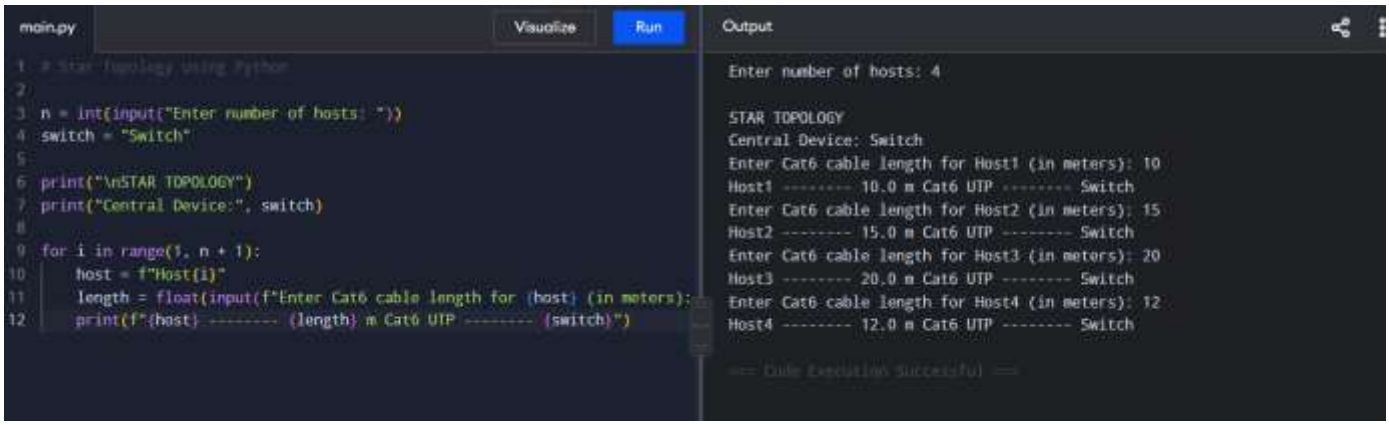
I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student*: M Nishanth**

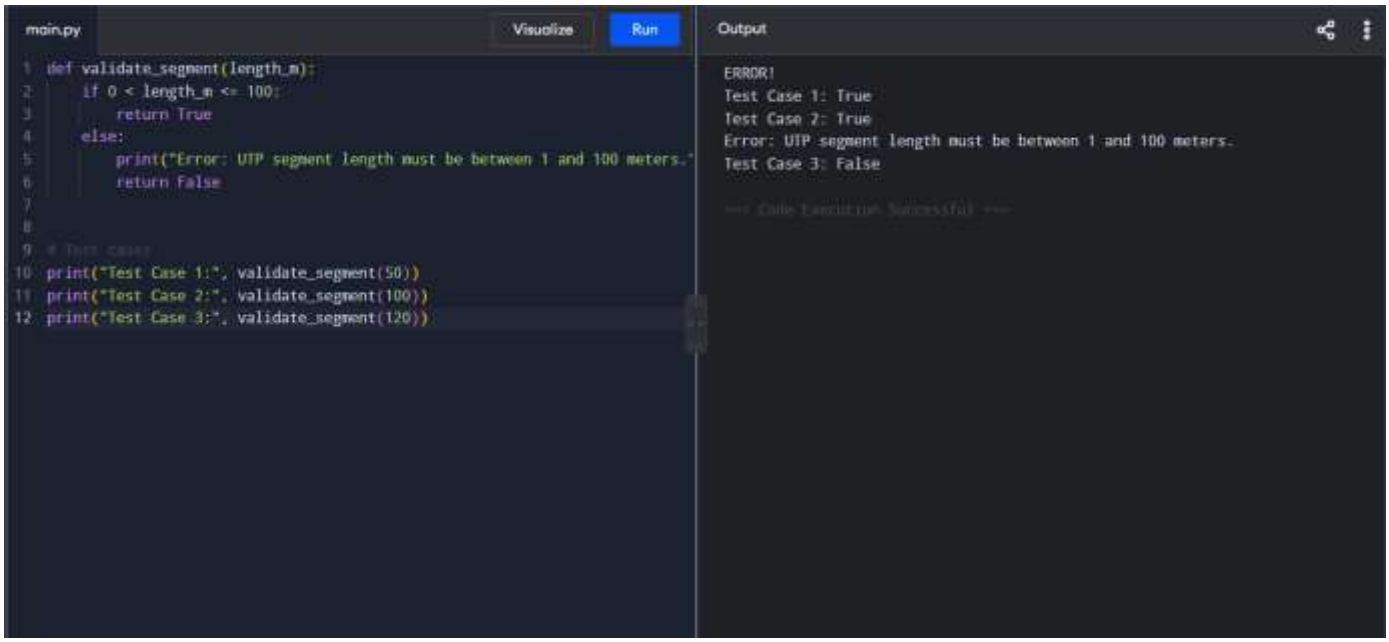
Annexure A

Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.



2. Code a function `validate_segment(length_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤100 m) and False with an error message otherwise. Include at least 3 test cases.



3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma_log.txt`.

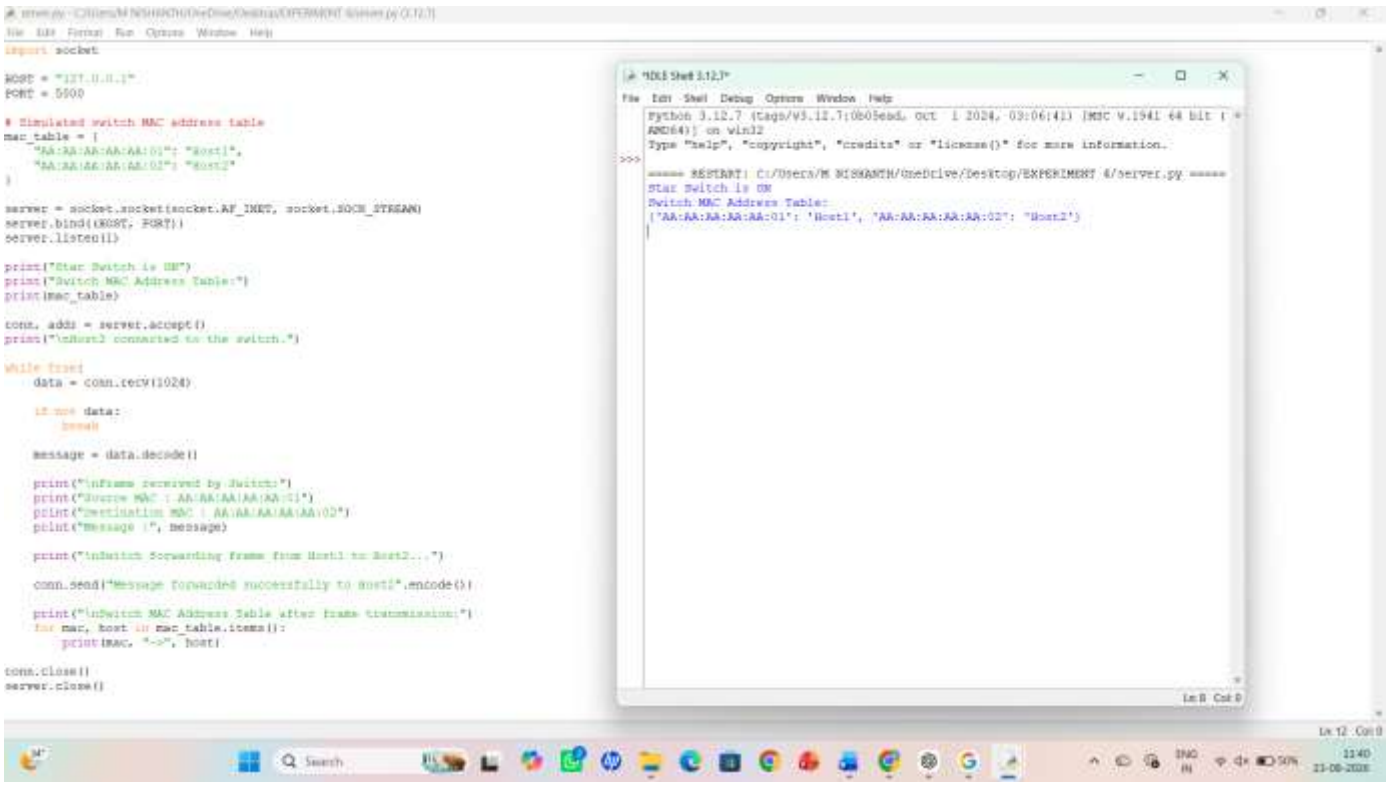
csma.py - C:/Users/M NISHANTH/Downloads/csma.py (3.12.7)

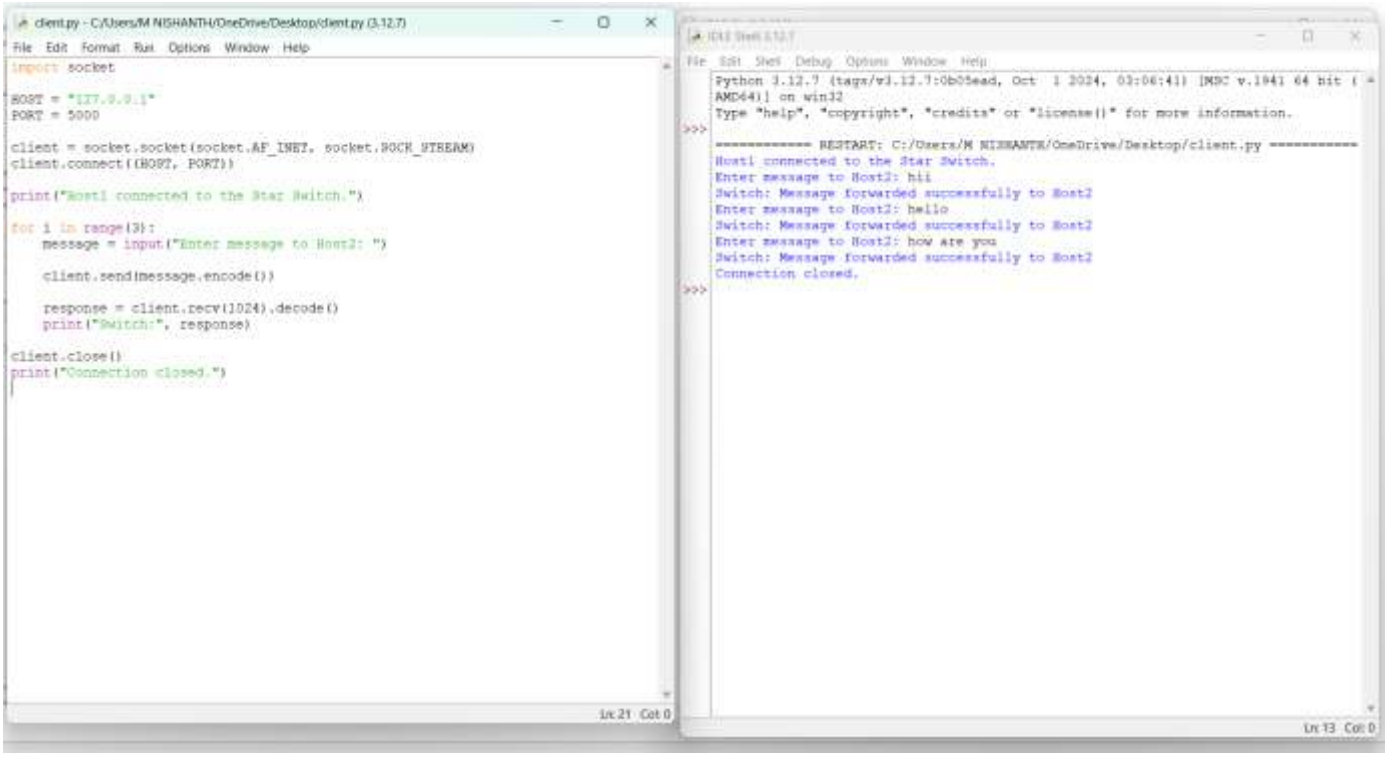
File Edit Format Run Options Window Help

```
import random
import time

hosts = ["Host1", "Host2", "Host3", "Host4"]

with open("csma_log.txt", "w") as log:
    log.write("CSMA/CD Simulation - Star Topology\n")
    log.write("-----------------------------------\n")

    for i in range(8):
        host = random.choice(hosts)

        # Randomly decide whether collision occurs
        collision = random.choice([True, False])

        if collision:
            backoff = random.randint(1, 10)

            message = (
                f"Collision detected by {host} | "
                f"Back-off interval = {backoff} ms"
            )

            print(message)
            log.write(message + "\n")

            time.sleep(backoff / 1000)

        else:
            message = f"{host} transmitted successfully | No collision"
            print(message)
            log.write(message + "\n")

print("\nSimulation completed.")
print("Log saved in csma_log.txt")
```

```
IDLE Shell 3.12.7
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.7 (tags/v3.12.7:0b05ead, Oct  1 2024, 03:06:41) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
================ RESTART: C:/Users/M NISHANTH/Downloads/csma.py ================
Collision detected by Host2 | Back-off interval = 5 ms
Collision detected by Host4 | Back-off interval = 4 ms
Host4 transmitted successfully | No collision
Host2 transmitted successfully | No collision
Collision detected by Host2 | Back-off interval = 10 ms
Host2 transmitted successfully | No collision
Host1 transmitted successfully | No collision
Host4 transmitted successfully | No collision

Simulation completed.
Log saved in csma_log.txt
>>>
```

4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.





client.py - C:/Users/M NISHANTH/OneDrive/Desktop/client.py (3.12.7)

File Edit Format Run Options Window Help

```
import socket

HOST = "127.0.0.1"
PORT = 5000

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))

print("Host1 connected to the Star Switch.")

for i in range(3):
    message = input("Enter message to Host2: ")

    client.send(message.encode())

    response = client.recv(1024).decode()
    print("Switch:", response)

client.close()
print("Connection closed.")
```

Ln: 21 Col: 0

IDLE Shell 3.12.7

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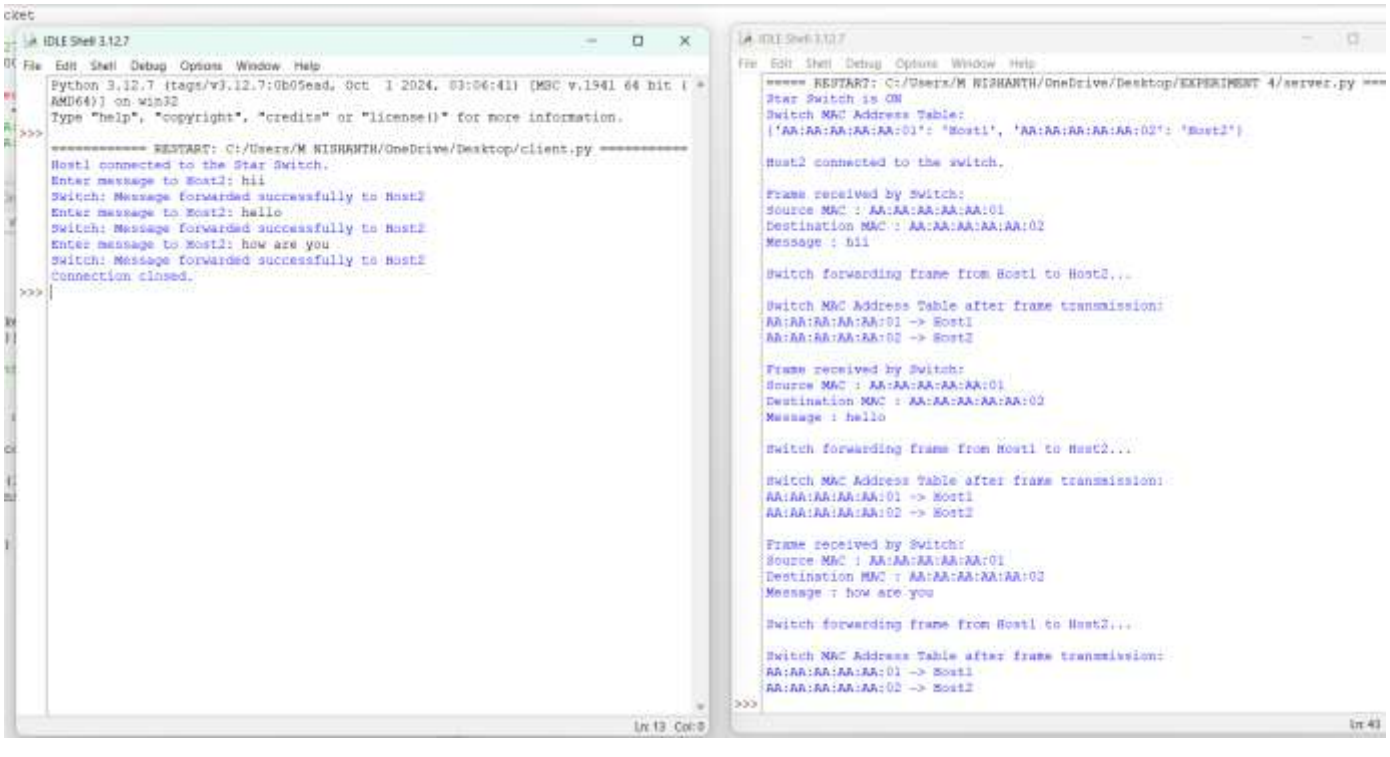
```
Python 3.12.7 (tags/v3.12.7:0b05ead, Oct  1 2024, 03:06:41) [MSC v.1941 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> ========== RESTART: C:/Users/M NISHANTH/OneDrive/Desktop/client.py ==========
Host1 connected to the Star Switch.
Enter message to Host2: hii
Switch: Message forwarded successfully to Host2
Enter message to Host2: hello
Switch: Message forwarded successfully to Host2
Enter message to Host2: how are you
Switch: Message forwarded successfully to Host2
Connection closed.
>>>
```

Ln: 13 Col: 0



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IDLE Shell 3.12.7

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```
Python 3.12.7 (tags/v3.12.7:0b05ead, Oct  1 2024, 03:06:41) [MSC v.1941 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> ========== RESTART: C:/Users/M NISHANTH/OneDrive/Desktop/client.py ==========
Host1 connected to the Star Switch.
Enter message to Host2: hii
Switch: Message forwarded successfully to Host2
Enter message to Host2: hello
Switch: Message forwarded successfully to Host2
Enter message to Host2: how are you
Switch: Message forwarded successfully to Host2
Connection closed.
>>>
```

Ln: 13 Col: 0

IDLE Shell 3.12.7

File Edit Shell Debug Options Window Help

```
===== RESTART: C:/Users/M NISHANTH/OneDrive/Desktop/EXPERIMENT 4/server.py ===
Star Switch is ON
Switch MAC Address Table:
{'AA:AA:AA:AA:AA:01': 'Host1', 'AA:AA:AA:AA:AA:02': 'Host2'}

Host2 connected to the switch.

Frame received by Switch:
Source MAC : AA:AA:AA:AA:AA:01
Destination MAC : AA:AA:AA:AA:AA:02
Message : hii

Switch forwarding frame from Host1 to Host2...

Switch MAC Address Table after frame transmission:
AA:AA:AA:AA:AA:01 -> Host1
AA:AA:AA:AA:AA:02 -> Host2

Frame received by Switch:
Source MAC : AA:AA:AA:AA:AA:01
Destination MAC : AA:AA:AA:AA:AA:02
Message : hello

Switch forwarding frame from Host1 to Host2...

Switch MAC Address Table after frame transmission:
AA:AA:AA:AA:AA:01 -> Host1
AA:AA:AA:AA:AA:02 -> Host2

Frame received by Switch:
Source MAC : AA:AA:AA:AA:AA:01
Destination MAC : AA:AA:AA:AA:AA:02
Message : how are you

Switch forwarding frame from Host1 to Host2...

Switch MAC Address Table after frame transmission:
AA:AA:AA:AA:AA:01 -> Host1
AA:AA:AA:AA:AA:02 -> Host2
>>>
```

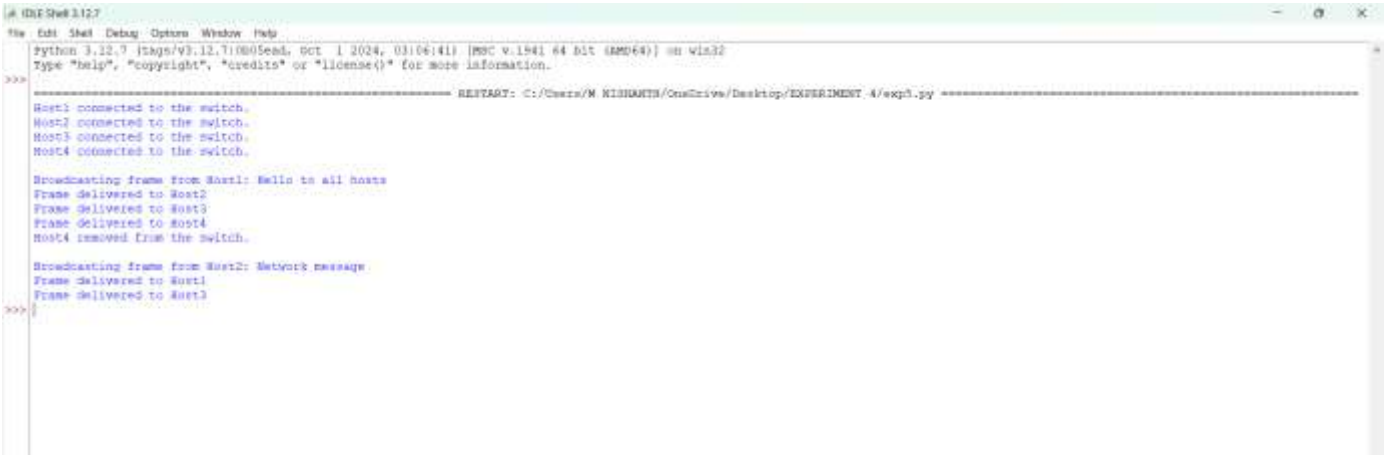
Ln: 43

5. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

exp5.py - C:/Users/M NISHANTH/OneDrive/Desktop/EXPERIMENT 4/exp5.py (3.12.7)

File Edit Format Run Options Window Help

```
        print(f"{host_id} connected to the switch.")

    def remove_port(self, host_id):
        if host_id in self.ports:
            self.ports.remove(host_id)
            print(f"{host_id} removed from the switch.")
        else:
            print(f"{host_id} is not connected.")

    def broadcast(self, frame):
        source = frame["source"]
        message = frame["message"]

        print(f"\nBroadcasting frame from {source}: {message}")

        for host in self.ports:
            if host != source:
                print(f"Frame delivered to {host}")


# Create a star switch
switch = StarSwitch()

# Add hosts
switch.add_port("Host1")
switch.add_port("Host2")
switch.add_port("Host3")
switch.add_port("Host4")

# Broadcast a frame
frame = {
    "source": "Host1",
    "message": "Hello to all hosts"
}

switch.broadcast(frame)

# Remove a host
switch.remove_port("Host4")

# Broadcast another frame
frame2 = {
    "source": "Host2",
    "message": "Network message"
}

switch.broadcast(frame2)
```


IDLE Shell 3.12.7
File Edit Shell Debug Options Window Help
Python 3.12.7 (tags/v3.12.7:0b05ead, Oct 1 2024, 03:06:41) [MSC v.1941 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/M NISHANTH/OneDrive/Desktop/EXPERIMENT 4/exp3.py ==========
Host1 connected to the switch.
Host2 connected to the switch.
Host3 connected to the switch.
Host4 connected to the switch.
Broadcasting frame from Host1: Hello to all hosts
Frame delivered to Host2
Frame delivered to Host3
Frame delivered to Host4
Host4 removed from the switch.
Broadcasting frame from Host2: Network message
Frame delivered to Host1
Frame delivered to Host3
>>>

1. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Understanding	15%	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
Algorithm Design	20%	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm
Code Quality	15%	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
Functionality	20%	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
Error Handling	10%	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
Efficiency	10%	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
Documentation & Comments	5%	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
Testing & Debugging	5%	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done